IJNRD.ORG ISSN : 2456-4184
INTERNATIONAL JOURNAL OF NOVEL RESEARCH AND DEVELOPMENT (IJNRD) | IJNRD.ORG
An International Open Access, Peer-reviewed, Refereed Journal

QUANTUM CRYPTOGRAPHY

Krish.S.Nagaral
UG Student,
School of Computer Science and Engineering,
Vellore Institute of Technology, Chennai Campus

Karnik Chauhan
UG Student
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai Campus

Yogendra Champa
UG Student
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai Campus

Akash Srikanth
UG Student
School of Computer Science and Engineering
Vellore Institute of Technology, Chennai Campus

Dr Balasaraswathi V R Professor,
School of Computer Science and Engineering (Scope),
Vellore Institute of Technology, Chennai Campus

Abstract:

In the rapidly evolving field of cyber security, integrating classical and quantum cryptographic techniques to protect sensitive information is becoming increasingly important. This project leverages the capabilities of Microsoft Azure to create a robust security framework that includes two servers: a host server and a user server. The host server uses quantum cryptography to generate encryption keys, while the user server implements the Fernet algorithm for data encryption.

Quantum cryptography, specifically Quantum Key Distribution (QKD), is used on the host server to generate keys that change every nanosecond. This constant key variation makes it virtually impossible for unauthorized parties to intercept and use the key for malicious purposes. Even if an attacker succeeds in intercepting a key, the rapid change in its value renders it useless almost immediately.

The Fernet algorithm, known for its simplicity and effectiveness, is used for the encryption and decryption processes. Fernet's reliance on AES encryption in conjunction with the dynamically changing, quantum-generated keys ensures a high level of security. The integration of quantum cryptography into the Fernet algorithm combines the real-time security of quantum key generation with the proven strength of AES encryption.

By leveraging the advanced infrastructure of Microsoft Azure, this project demonstrates how quantum cryptography and classical encryption can be seamlessly integrated to create a highly secure communication environment. This approach not only protects against classical threats but also future-proofs the system against potential quantum attacks and ensures the confidentiality and integrity of data in an increasingly complex digital landscape.

Keywords: Quantum key distribution (QKD), lattice-based cryptography, hybrid cryptographic systems, Azure Bastion Service, quantum mechanics, superposition, quantum bits (qubits), BB84 protocol, Ekert's E91 protocol, decoherence, cross-platform compatibility.

INTRODUCTION

Quantum cryptography uses the smallest discrete units of physical properties known as quanta and is based on the principles of quantum mechanics to secure communications. Quantum mechanics introduces unique concepts such as superposition and entanglement that differ significantly from classical physics.

In the context of cryptography, these principles are used to ensure security through physical laws rather than mathematical complexity. Key applications include quantum key distribution (QKD) protocols such as BB84 and Ekert's E91, which use quantum bits (qubits) to enable secure key distribution. These qubits can exist in multiple states simultaneously, and any measurement interferes with them, revealing eavesdropping attempts. Entanglement-based protocols further enhance security by establishing correlations between shared qubits, where each interception changes its state, enabling detection.

Practical implementations of QKD have been demonstrated with fiber optic cables and satellite-based systems such as the Chinese Micius satellite, which enable secure key distribution over long distances. Quantum error correction and repeaters overcome challenges such as photon loss and decoherence and extend the range and robustness of quantum communication.

The advantages of quantum cryptography include unconditional security based on the no-cloning theorem and the uncertainty principle, which ensures that any observation interferes with the system and makes eavesdropping detectable. It is also believed to be secure against future attacks on quantum computers.

However, challenges such as photon loss, decoherence and the need for specialized hardware and infrastructure make it more costly and complex than classical cryptographic systems.

Despite these challenges, ongoing research aims to standardize protocols and develop cost-effective solutions for widespread application. The reviewed papers provide a comprehensive understanding of the theoretical foundations, practical implementations and advances in quantum cryptography and demonstrate its potential to revolutionize secure communication.

THE RATIONALE BEHIND THE FERNET ALGORITHM

Fernet is a symmetric encryption method that ensures the confidentiality and integrity of messages and is therefore suitable for use in quantum cryptography. It provides robust security through AES encryption in CBC mode with a 128-bit key and HMAC for integrity checks, protecting against common cryptographic attacks.

Fernet's simplicity, built-in timestamping to prevent replay attacks and cross-platform compatibility make it accessible and versatile for developers. Although Fernet is not

inherently quantum resistant, it can be integrated into a layered security approach where quantum key distribution (QKD) handles the key exchange and Fernet secures the payload.

This combination increases the overall robustness of the communication channels. Alternatives such as AES-GCM and Salsa20/ChaCha20 offer similar encryption and integrity benefits with high performance and efficiency, making them suitable for integration with quantum-safe key exchange mechanisms.

New quantum-safe algorithms such as lattice-based cryptography provide direct resistance to quantum attacks, ensuring long-term security. Fernet's strengths in encryption and integrity combined with its ease of use make it a valuable tool in the hybrid cryptographic systems that will support secure communications in the quantum age.

LITERATURE SURVEY

In [3] Quantum cryptography could well be the first application of quantum mechanics at the single quantum level. The rapid progress that has been made in recent years, both in theory and experiment, is discussed, with a focus on open questions and technological problems.

[4] After a brief introduction to classical cryptography, we explain in detail how quantum cryptography works. We then present an elegant experimental implementation based on a self-balancing interferometer with Faraday mirrors. This phase encoding setup requires neither alignment of the interferometer nor polarization control and therefore greatly facilitates the experiment. It is also characterized by excellent fringe visibility. Next, we estimate the practical limits of quantum cryptography. The importance of detector noise is illustrated and ways to reduce it are presented. With today's technologies, maximum distances of about 70 km with bit rates of 100 Hz are achievable.

In [5], quantum cryptography makes use of special quantum properties of nature to fulfil a specific cryptographic task. Most quantum cryptographic protocols are — at least in principle— - information-theoretically secure, which is a very strong notion of security since it is derived purely from information theory.

In [6], BBN, Harvard, and Boston University are building the DARPA Quantum Network, the world's first network to provide end-to-end network security via high-speed quantum key distribution, and testing this network against sophisticated eavesdropping attacks. The first network connection has been in operation in our laboratory since December 2002 and is constantly being further developed.

It provides a virtual private network between private enclaves, with user traffic protected by a weakly coherent implementation of quantum cryptography. This prototype is suitable for use in metropolitan areas over standard telecommunication fibres (dark fibre). In this paper, we introduce quantum cryptography, discuss its relationship to modern secure networks, and describe its unusual physical layer, its specialised quantum cryptography protocol suite (which is quite interesting in its own right), and our extensions to IPsec to integrate it with quantum cryptography.

In [7], quantum cryptography utilises principles of quantum physics for the secure processing of information. A well-known example is secure communication, i.e. the task of transmitting confidential messages from one place to another. The cryptographic requirement here is that the transmitted messages remain inaccessible to anyone other than the intended recipient, even if the communication channel is not trustworthy. In classical cryptography, this can usually only be guaranteed under the assumption of computational hardness, e.g. if the factorization of large integers is not possible. In contrast, the security of quantum cryptography is based entirely on the laws of quantum mechanics. Here, this physical notion of security is reviewed, with a focus on quantum key distribution and secure communication.

In [8] quantum cryptography is a new method for secret communication that provides the ultimate security guarantee of inviolability of a law of nature. In this paper, we describe the theory of quantum cryptography, its potential importance, and the development of a prototype system at Los Alamos that uses the phenomenon of single photon interference to perform quantum cryptography over a fibre optic communication link.

In [9] quantum cryptography is the art and science of exploiting quantum mechanical effects to accomplish cryptographic tasks. The most well-known example of this discipline is

quantum key distribution (QKD), but there are many other applications such as quantum money, randomization, secure two-party and multi-party computation, and delegated quantum computation. Quantum cryptography also addresses the limitations and challenges posed by quantum adversaries. These include the impossibility of quantum bit commitment, the difficulty of quantum rewinding, and the definition of quantum security models for classical primitives. In this survey article, aimed primarily at cryptographers unfamiliar with the quantum world, we provide an overview of the field of theoretical quantum cryptography, focusing on the constructions and constraints that go beyond the realm of QKD.

In [11] We live in a quantum world – something that physicists have regarded with astonishment for more than seventy years. However, we now know that quantum physics not only represents a radical departure from classical physics, but also offers many new possibilities for information processing.

In [12], quantum cryptography makes use of the quantum mechanical behavior of nature for the design and analysis of cryptographic procedures. Ideally (but not always), quantum cryptography enables the development of cryptographic systems whose security is guaranteed by the laws of nature alone. This is in stark contrast to standard cryptographic methods, which can in principle be cracked, i.e. if sufficient computing power is available.

From a theoretical point of view, quantum cryptography offers a nice interplay between the mathematics of adversarial behaviour and quantum information theory.

In this review article, we discuss the traditional application of quantum cryptography, quantum key distribution (QKD), from a modern perspective and discuss some recent developments related to quantum two-party cooperation (2PC). QKD allows two remote parties to communicate in a provably secure manner in the presence of an external eavesdropper, while 2PC is about protecting information from potentially malicious insiders. We show the basic idea of constructing quantum cryptographic schemes, but we also show some connections to quantum information theory that are needed for the rigorous security analyses, and we discuss some of the relevant quantum information theory results.

METHODOLOGY

In this project, we use the capabilities of Microsoft Azure to implement a highly secure communication system that combines quantum cryptography with the Fernet encryption algorithm. Our infrastructure consists of two servers: a host server and a user server, both connected via the Azure service Bastion. This provides a secure and seamless RDP and SSH connection to virtual machines without exposing them to the public internet. This eliminates the risk of external threats attempting to exploit the exposed endpoints and ensures that our communication system remains protected from unauthorized access.

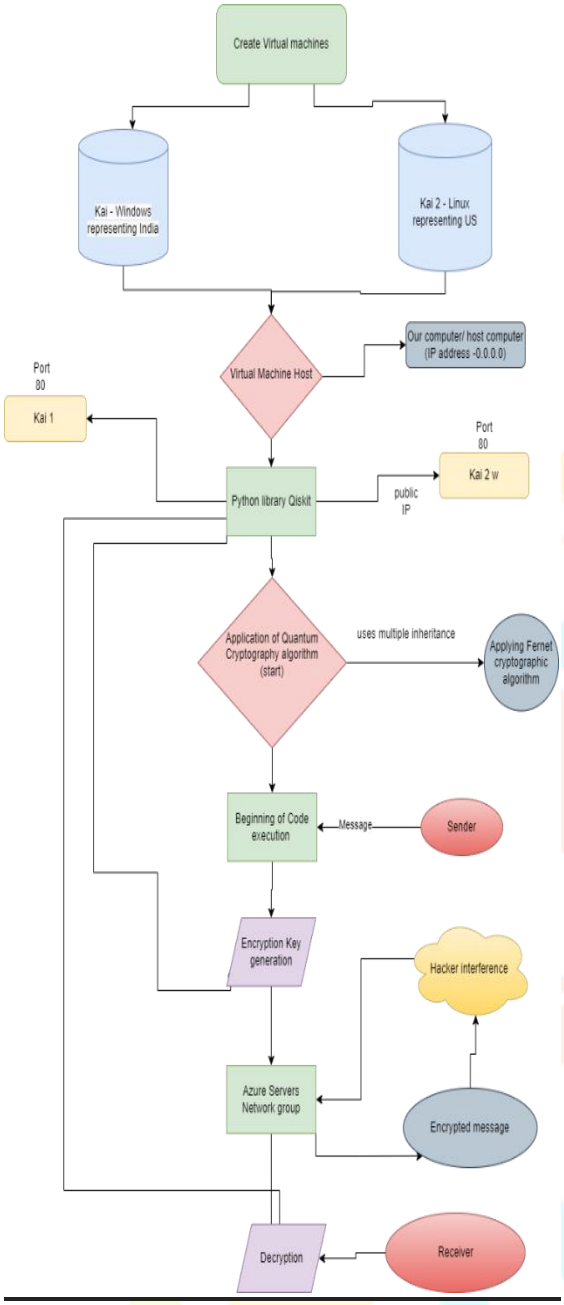
The host server is equipped with quantum cryptographic capabilities to generate encryption keys using advanced quantum technology. This technology ensures that the keys are generated with such complexity and rapid variability that it is virtually impossible to intercept or replicate them. The only reason for this is that the keys change every nanosecond. This means that even if an attacker were to capture a key, it would be out of date by the time they could use it. This rapid key rotation takes advantage of the inherent unpredictability and security of quantum mechanics to significantly improve data protection.

For encryption, we use the Fernet algorithm, which is known for its strong security features, including AES encryption in CBC mode with a 128-bit key and HMAC for integrity checks. The quantum-generated keys are used together with Fernet to encrypt data to ensure that the data remains confidential and tamper-proof. The integration of Fernet with quantum key distribution provides a robust encryption mechanism that benefits from both classical and quantum security measures.

This project is an example of a hybrid approach to cryptographic security, combining the strengths of quantum key distribution with the proven effectiveness of classical encryption algorithms such as Fernet. The use of rapidly changing quantum keys ensures that any potential interception is hopeless, while Azure Bastion's secure connectivity minimizes the attack surface. Together, these technologies create a fortified environment for secure data transfer between the host and user servers, making it an ideal solution for applications that

require a high level of security and data integrity.

DATA DEPICTION



NOTE ON UNDERLYING CONCEPT OF QUANTUM COMPUTING

In quantum computing, the spin of the electrons is very important because it is something like the quantum version of a bit, a qubit. A qubit can assume two states, like 0 and 1 in classical computing, but it can also assume a superposition of these states, i.e. it can assume both states simultaneously. This superposition enables quantum computers to do far more than classical computers.

To control the spin of the electrons and the qubits they represent, we use magnetic fields or so-called spin-orbit coupling. This allows us to manipulate the qubits and do all sorts of fancy things, such as process information much faster and do much more complex things than we could with normal computers. So quantum computers are a big deal, allowing us to do things we never thought possible before, all thanks to the spin of these electrons.

CONCLUSION

This project has successfully demonstrated the powerful synergy between quantum cryptography and cloud computing by leveraging the advanced capabilities of Microsoft Azure to create a highly secure communication environment. By integrating quantum-generated keys with the Fernet encryption algorithm, the system ensures dynamic and robust data protection. The host server uses quantum technology to generate keys that change every nanosecond, making unauthorized interception and decryption virtually impossible.

The implementation of this dual server setup on Azure not only underlines the feasibility of combining quantum cryptographic techniques with classical encryption methods, but also sets a new standard for cybersecurity. This innovative approach ensures continuous, adaptive security capable of outperforming potential threats and protecting sensitive information in real time.

Overall, this project underlines the immense potential of quantum cryptography in improving data security. It paves the way for future advances and sets a precedent for secure communication in the digital age.

REFERENCE

[1] Bennett, C. H., Brassard, G., & Ekert, A. K. (1992). Quantum Cryptography. Scientific American, 267(4), 50–57.

[2] Dušek, M., Lütkenhaus, N., & Hendrych, M. (2006). Quantum cryptography. *Progress in optics*, *49*, 381-454.

[3] Gisin, N., Ribordy, G., Tittel, W., & Zbinden, H. (2002). Quantum cryptography. *Reviews of modern physics*, *74*(1), 145.

[4] Zbinden, H., Bechmann-Pasquinucci, H., Gisin, N., & Ribordy, G. (1998). Quantum cryptography. Applied Physics B: Lasers & Optics, 67(6).

[5] Grasselli, F. (2021). Quantum Cryptography: From Key Distribution to Conference Key Agreement. Springer.

[6] Elliott, C., Pearson, D., & Troxel, G. (2003, August). Quantum cryptography in practice. In Proceedings of the 2003 conference on Applications, technologies, architectures, and protocols for computer communications (pp. 227-238).

[7] Portmann, C., & Renner, R. (2022). Security in quantum cryptography. Reviews of Modern Physics, 94(2), 025008.

[8] Hughes, R. J., Alde, D. M., Dyer, P., Luther, G. G., Morgan, G. L., & Schauer, M. (1995). Quantum cryptography. Contemporary Physics, 36(3), 149-163.

[9] Broadbent, A., & Schaffner, C. (2016). Quantum cryptography beyond quantum key distribution. Designs, Codes and Cryptography, 78, 351-382.

[10] Lo, H. K., & Zhao, Y. (2008). Quantum cryptography. arXiv preprint arXiv:0803.2507.

[11] Gisin, N., Ribordy, G., Tittel, W., & Zbinden, H. (2002). Quantum cryptography. Reviews of modern physics, 74(1), 145.

[12] Fehr, S. (2010). Quantum cryptography. Foundations of Physics, 40, 494-531.

International Journal Research Journal
IJNRD
Research Through Innovation